DISCRETE STRUCTURES

CMSC250

Chapter 12: Modular Arithmetic

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Remainders, congruence classes, and proof techniques



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1 What modular arithmetic actually means

Ordinary equality asks whether two integers are the same. Congruence modulo m asks whether they leave the same remainder when divided by m. For integers a, b and positive integer m,

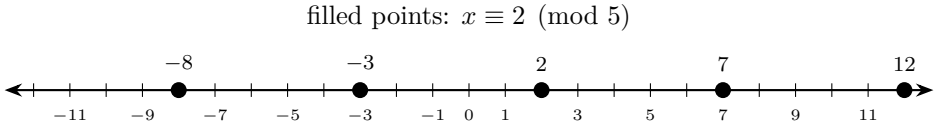
$$\boxed{a \equiv b \pmod{m} \iff m \mid (a-b).}$$

This is the *definition*. Equivalently, there is an integer k such that $a - b = mk$. For example $17 \equiv 2 \pmod 5$ because $17 - 2 = 15 = 5(3)$.

Key idea. The symbol $\equiv$ means congruent, not equal. The integers 17 and 2 are different, but have the same remainder modulo 5.

2 A number line showing congruence classes

Numbers congruent to 2 modulo 5 occur every five units: $\ldots, -8, -3, 2, 7, 12, \ldots$. All belong to the same residue class.



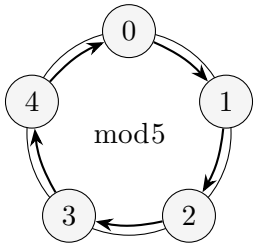
Worked example: positive and negative residues

Question. Find the least nonnegative remainder of -17 modulo 5.

Solution. We want $-17 = 5q + r$ with $0 \le r < 5$. Since $-17 = 5(-4) + 3$, the remainder is 3, so $-17 \equiv 3 \pmod 5$. Check directly: $-17 - 3 = -20 = 5(-4)$.

3 Residue classes and the clock picture

Modulo 5, every integer is congruent to exactly one of $0, 1, 2, 3, 4$. After 4, the residues return to 0.



For example $4 + 3 = 7 \equiv 2 \pmod 5$. We can imagine moving three positions clockwise from residue 4 to residue 2.

4 Proving congruence from the definition

Worked example: a direct congruence proof

Question. For all integers a, b and positive integers m, if $a \equiv b \pmod{m}$, then $a + 3m \equiv b \pmod{m}$.

Proof. Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}_{>0}$ be arbitrary. Assume $a \equiv b \pmod{m}$. We will show $a + 3m \equiv b \pmod{m}$. By the definition of congruence, $a - b = mk$ for some $k \in \mathbb{Z}$. Therefore

$$(a + 3m) - b = (a - b) + 3m = mk + 3m = m(k + 3).$$

Since $k + 3 \in \mathbb{Z}$ by closure, $m \mid ((a + 3m) - b)$. Hence $a + 3m \equiv b \pmod{m}$ by definition. ■

5 Reflexive, symmetric, and transitive

Congruence modulo a positive integer is an equivalence relation. Each property follows from the definition.

Worked example: prove symmetry

Question. If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$.

Proof. Let $a, b \in \mathbb{Z}$ and $m > 0$ be arbitrary. Assume $a \equiv b \pmod{m}$. Then $a - b = mk$ for some integer k. Hence $b - a = m(-k)$. Since $-k \in \mathbb{Z}$, $m \mid (b - a)$, and so $b \equiv a \pmod{m}$. ■

Worked example: prove transitivity

Question. If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.

Proof. Let $a, b, c \in \mathbb{Z}$ and $m > 0$ be arbitrary. Suppose $a - b = mr$ and $b - c = ms$ for some $r, s \in \mathbb{Z}$. Adding yields $a - c = m(r + s)$. Closure gives $r + s \in \mathbb{Z}$. Therefore $a \equiv c \pmod{m}$. ■

6 Addition, subtraction, and multiplication rules

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a + c \equiv b + d \pmod{m}, \qquad a - c \equiv b - d \pmod{m}, \qquad ac \equiv bd \pmod{m}.$$

Worked example: why multiplication is valid

Question. Prove that $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ imply $ac \equiv bd \pmod{m}$.

Proof. Let the integers and positive modulus be arbitrary, and assume both congruences. Then $a - b = mr$ and $c - d = ms$ for integers r, s. Observe

$$ac - bd = c(a - b) + b(c - d) = m(cr + bs).$$

Since $cr + bs \in \mathbb{Z}$ by closure, $m \mid (ac - bd)$. Thus $ac \equiv bd \pmod{m}$. ■

7 Powers and reducing large calculations

If $a \equiv b \pmod{m}$, then $a^n \equiv b^n \pmod{m}$ for every nonnegative integer n. This follows from repeated multiplication.

Worked example: find the remainder of a large power

Question. Find 7^{20} mod 6.

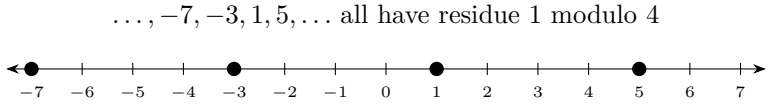
Solution. Since $7 \equiv 1 \pmod{6}$, we have $7^{20} \equiv 1^{20} = 1 \pmod{6}$. The remainder is $\boxed{1}$.

Worked example: a repeating pattern of powers

Question. Find the remainder when 3^{25} is divided by 7.

Solution. Calculate $3^1 \equiv 3$, $3^2 \equiv 2$, $3^3 \equiv 6$, $3^4 \equiv 4$, $3^5 \equiv 5$, and $3^6 \equiv 1 \pmod{7}$. Thus powers repeat in blocks of six. Since $25 = 6(4) + 1$, $3^{25} \equiv 3^1 \equiv 3 \pmod{7}$.

8 Congruence classes as repeating positions



A residue class is an infinite set, not just one remainder. We often choose $0, 1, \ldots, m-1$ as *representatives* of the m classes.

9 When division and cancellation fail

You may add, subtract, and multiply congruences, but you cannot always divide both sides by the same number. For instance $2 \cdot 1 \equiv 2 \cdot 4 \pmod{6}$ because $2 \equiv 8 \pmod{6}$, yet $1 \not\equiv 4 \pmod{6}$.

Key idea. Cancellation of a factor c modulo m is valid when $\gcd(c, m) = 1$. Without that condition, cancellation may change the modulus or fail entirely.

Worked example: solve a linear congruence by checking residues

Question. Find all residues $x \in \{0, 1, 2, 3, 4, 5\}$ satisfying $2x \equiv 4 \pmod{6}$.

Solution. The possibilities are $x = 2$ and $x = 5$, because $2(2) = 4$ and $2(5) = 10 \equiv 4 \pmod{6}$. Notice that simply dividing by 2 and reporting only $x = 2$ would miss $x = 5$.

10 Connecting modular arithmetic to divisibility

Worked example: a divisibility proof using remainders

Question. Show that for every integer n, $n^3 - n$ is divisible by 3.

Proof. Let $n \in \mathbb{Z}$ be arbitrary. Every integer has exactly one of the residues $0, 1, 2$ modulo 3. If $n \equiv 0$, then $n^3 - n \equiv 0^3 - 0 = 0$. If $n \equiv 1$, then $n^3 - n \equiv 1^3 - 1 = 0$. If $n \equiv 2$, then $n^3 - n \equiv 2^3 - 2 = 6 \equiv 0 \pmod 3$. Thus in every possible case $n^3 - n \equiv 0 \pmod 3$, meaning $3 \mid (n^3 - n)$. Since n was arbitrary, the claim holds. ■

11 Common mistakes

- Confusing $a \equiv b \pmod m$ with $a = b$.
- Treating a negative remainder as the canonical representative when $m > 0$.
- Cancelling a common factor without checking whether it is relatively prime to m.
- Writing a modular proof without exhibiting $a - b = mk$ or citing a valid congruence rule.
- Assuming a few numerical examples prove a universal statement.

12 Practice questions

1. Find the least nonnegative residues of -29 modulo 6 and 53 modulo 8.
2. Prove: if $a \equiv b \pmod m$, then $4a + 7 \equiv 4b + 7 \pmod m$.
3. Find $11^{17} \bmod 5$.
4. Find every $x \in \{0, 1, 2, 3, 4, 5, 6, 7\}$ such that $2x \equiv 6 \pmod 8$.
5. Prove: if $a \equiv b \pmod m$, then $a^2 \equiv b^2 \pmod m$ directly from divisibility.
6. Show that $n^2 \equiv 0$ or $1 \pmod 4$ for every integer n.

13 Fully worked practice solutions

1. $-29 = 6(-5) + 1$, so the residue is 1. Also $53 = 8(6) + 5$, so the residue is 5.
2. **Proof.** Let $a, b \in \mathbb{Z}$ and $m > 0$ be arbitrary. If $a \equiv b \pmod m$, then $a - b = mk$ for some $k \in \mathbb{Z}$. Hence $(4a + 7) - (4b + 7) = 4mk = m(4k)$. Closure gives $4k \in \mathbb{Z}$, so the required congruence holds. ■
3. $11 \equiv 1 \pmod 5$, so $11^{17} \equiv 1^{17} = 1 \pmod 5$.
4. Test residues or rewrite $2x - 6 = 8k$. The solutions are $x = 3$ and $x = 7$.
5. **Proof.** Since $a - b = mk$ for some integer k, we have $a^2 - b^2 = (a - b)(a + b) = m[k(a + b)]$. Closure gives $k(a + b) \in \mathbb{Z}$, so $a^2 \equiv b^2 \pmod m$. ■

6. **Proof by cases.** If n is even, $n = 2k$ and $n^2 = 4k^2 \equiv 0 \pmod 4$. If n is odd, $n = 2k + 1$ and $n^2 = 4(k^2 + k) + 1 \equiv 1 \pmod 4$. Every integer is even or odd, so the result follows. ■